KEONWOO KIM

AI Engineer - adopts LLM tooling early, measures cost and latency in production, cuts what doesn't pay off.

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WORK EXPERIENCE

Polyfact — AI Software Engineer Aug 2024 – Sep 2025 · Paris, France

B2B policy-intelligence platform on French and EU parliamentary activity. The senior engineers left soon after I joined; from then on I carried the product, backend, AI pipeline and SvelteKit frontend, at times alone with the CEO, and reviewed and trained the engineers and interns who came after me.

Python · FastAPI · TypeScript (Deno, Node) · SvelteKit · Supabase/Postgres · Turbopuffer · Pinecone · OpenAI · Sentry · OpenTelemetry · GCP · LangChain/LangGraph/LangSmith

- **Built the AI backend from an empty repo (FastAPI → Deno → Node on GCP):** the service every later AI feature ran on. Per-request token cost logged from the first version; blocking I/O rewritten async so generation requests ran concurrently instead of queueing; 100-page document structuring moved to streaming with live progress.
- **Vector search and RAG introduced to the product**: profile-to-activity matching on Pinecone, then a Turbopuffer index for the amendments chat built around the store's limits (French BM25 tokenizer, 4KB attribute cap, per-document failure isolation), hybrid retrieval and reranking on top.
- **LangChain, LangGraph and LangSmith introduced for the chat and report agents, then owned**: every new model and API capability validated on our own outputs before it shipped, usually within days of release.
- **Documentation the team coded against**: an error-handling architecture (RFC 7807, one log per failure), observability guides in English and Korean, a rules file so AI-assisted code followed the same conventions, a per-owner TODO rollout so a shared package migrated in parallel through small PRs. Most of the team's PRs reviewed.
- **Operations nobody else had time for**: Sentry across four services through the shared logger with zero call-site changes, OpenTelemetry on the video pipeline, a daily digest email; three Supabase projects merged into one, Svelte 5, the AI backend moved from Deno to Node into the monorepo with history preserved.

Intern to permanent contract in six months. CEO reference, Dec 2024: "top 1%" in conscientiousness; "his issues, PR comments and technical notes are the most organized and consistent"; "I would bet on him."

Belage — Freelance Full-Stack Engineer Oct 2025 – Sep 2026 · Paris, France

A multi-brand studio ran bookings, consultations, payments and invoicing by hand. I built and run the replacement alone, customer site to back office.

Next.js · TypeScript · Supabase/Postgres (RLS, PL/pgSQL, pg_cron) · SumUp (online + card-present) · Google Calendar API · Vercel · pgTAP

- **Customer site and back office** on one Postgres schema serving every brand: portfolios, booking, payment, reviews and blog for clients; scheduling, staff roles and invoices for the studio. Booking, payment transitions and invoicing run as PL/pgSQL RPCs in single transactions behind RLS and server-side role checks.
- **Payments end to end:** online and in-studio card payments on one atomic settlement path, idempotent webhooks re-verified against the provider, a reconciliation cron for payments the browser lost, credit notes only after the provider confirms settlement.
- **Invoicing and accounting automated:** French statutory facture/avoir numbering (CGI art. 242 nonies A) on an atomic counter with an independent database constraint, PDF generated and emailed on settlement, Korean NTS cash-receipt export, reminders on pg_cron.
- **The studio's daily work automated:** consultations on KakaoTalk, WhatsApp, Instagram and email as stage-triggered messages, two-way Google Calendar sync per photographer, six languages from one source, and everything from prices to templates to brands editable from the back office, so the studio extends the system without a developer.

ADDITIONAL EXPERIENCE *Products I build and run*

Bonjour Admin — AI Software Engineer & Founder Feb 2026 – Present · paying subscribers

Every foreigner in France is a different case. General chatbots give general answers; Bonjour Admin answers for the person asking, from French law and official guidance only, with figures and references verified. Six languages. Built, run and monetised alone, company included.

Next.js 16 / React 19 · TypeScript · Supabase + pgvector · OpenAI Responses API · Cohere rerank · Upstash Redis · Polar.sh · PostHog · Vercel

- **Personalised retrieval:** a profile (nationality, region, permit status, family, work) steers retrieval through tags and RRF boosts and picks the applicable legal framework; users attach their own immigration-portal data, parsed against 110+ portal endpoints I mapped by reverse-engineering.
- **Trusted sources only:** six legal codes synced daily from Légifrance, fifteen government sources recrawled weekly, web search on a 55-domain whitelist, statutory figures versioned and injected as ground truth.

- **Retrieval as a pipeline, answering as an agent:** retrieval gate → semantic cache → weighted RRF over full-text and dual embeddings (pgvector) → Cohere rerank → quality gate; three tool-calling rounds over ten functions; three parallel verifiers (CRAG, legal-reference existence, citation entailment). LLM-generated SQL runs as an anonymous role that can read four public fact tables and nothing else.
- **Cost, latency and quality measured, then cut:** per-stage cost telemetry showed verification at 80% of spend and web pre-search at 49% of a legal answer, so pre-search went stale-only; search 18 s → 0.9 s; the retrieval gate skips 60–70% of calls on follow-ups; a 100+ case eval set with six LLM-as-judge metrics runs weekly in CI.
- **The whole company, alone:** fake-door test before the pipeline, Polar subscriptions with per-feature quotas, 1,000+ programmatic SEO pages, PostHog funnels, GDPR- and EU AI Act-compliant terms in three languages, INPI-registered trademark; around the chat, deadline reminders, a 494-question citizenship-exam trainer, document analysis and community experiences.

SonnanAI — AI Software Engineer Jan 2026 – Present · Paris, France

Real-time Korean → French interpretation for a Paris church: live subtitles and speech.

Python · Flask (SSE) · Silero VAD · OpenAI transcribe + local Whisper · Qwen2.5-7B LoRA · Ollama · pytest

- **Streaming STT → translation → TTS pipeline with graceful degradation:** VAD segmentation, cloud STT with local Whisper fallback, three-tier translation (weekly cache → local LoRA → cloud), request hedging under an 8 s wall clock, a circuit breaker on cloud STT, a fully offline path.
- **Qwen2.5-7B LoRA fine-tuned on a self-built corpus:** three years of bilingual sermon archives normalised and sentence-aligned into ~8K pairs, served locally through Ollama; a distillation flywheel turns every live session into context-aware retraining pairs, with train/eval leakage blocked.
- **Guards calibrated on real services:** an STT hallucination gate set from measured logprob distributions (speech above -0.12, phantoms below -1.3); a weekly pre-translation cache that absorbs 18.5% of utterances.

EDUCATION

École 42 (Paris) — IT Architecture Expert, RNCP 7 (Master's equivalent) Oct 2023 – Oct 2026 · Paris, France

Data and database architecture option. Project-only and peer-evaluated; 28 projects, core ML from scratch in numpy.

AI / ML

- **numpy-only MLP:** backpropagation derived by hand, Adam and L2 on top; gradient check to 1.72e-10.
- **EEG brain-computer interface:** motor-imagery CSP written from scratch as a generalised eigenvalue problem over two class covariances, matching SciPy to 1e-15; a test-data leak fixed inside the cross-validation fold.
- **CNN leaf-disease classifier (PyTorch):** 99.79% held-out on 8 classes, matching a fine-tuned EfficientNet-B0.
- **Also:** a Q-learning snake that generalises to board sizes it never saw; linear and logistic regression.

SYSTEMS, SECURITY, BLOCKCHAIN

- **x86 kernel from scratch (C, NASM):** higher-half paging with a recursive page directory, frame allocator from the Multiboot memory map, kmalloc/vmalloc heaps, no libc; regression-tested headlessly via the QEMU monitor.
- **Rust Gomoku engine:** Negamax, null-move pruning, transposition table, Lazy SMP; depth 10-17 under 500 ms.
- **Also:** GitOps cluster (K3s, K3d, ArgoCD); server-authoritative multiplayer Tetris; fourteen OWASP Top 10 exploits by hand; a BEP-20 token with 2-of-3 multisig and a fully on-chain SVG NFT; eight Flutter apps.

École 42 (Seoul) Mar 2021 – May 2023 · Seoul, South Korea

- **Grand Prize, 42 Hackathon (Minister of Science and ICT Award):** designed a BMP image-processing teaching project in pure C with a reference implementation and autograder; presented at 42 Amsterdam, Brussels and Paris.
- **HTTP/1.1 server in C++98** on a non-blocking kqueue event loop: static files, uploads and CGI.
- **Also:** libc, a bash-subset shell, dining philosophers and a raycasting FPS engine (C); STL containers (C++98); a hardened Debian VM and Docker stack; multiplayer Pong with OAuth and 2FA (NestJS + React).

SKILLS

- **Proficient —** TypeScript, Python, C / C++, SQL (PostgreSQL)
- **Experienced with —** Rust, Deno, Solidity, Dart / Flutter, x86 assembly
- **AI systems —** RAG (hybrid retrieval, RRF, reranking), agentic tool loops, CRAG, evals (LLM-as-judge), LLM cost and latency engineering, output sandboxing, LoRA fine-tuning, local serving (Ollama), speech pipelines (VAD, STT, TTS)
- **Backend & Web —** Next.js, FastAPI, Flask, SvelteKit, Node / Deno, Supabase/Postgres (RLS, PL/pgSQL), Redis, SSE / WebSocket
- **Infrastructure —** GCP Cloud Run, Vercel, Docker, K3s + ArgoCD, GitHub Actions, Sentry, OpenTelemetry, PostHog
- **Languages —** Korean (native) · English (professional) · French (basic)
- **Open source —** CUBRID (2021): Double Write Buffer bottleneck profiled, optimisation patch contributed upstream.